

The Craft of Computing

Course Overview

Outline

- What is this course about?
- Overview of course policies

What is this course about?

- First course in the Information Sciences major and minor
- Hands-on exploration of computing across three facets:
 - Command line interface
 - Python programming
 - Microcontrollers (ESP32)
- Goal: move from passive tech consumers to active creators and problem solvers
- Emphasis on teamwork, self-directed learning, and getting to know classmates

Course Objectives

After completing the course, students should be able to:

- **Execute** basic computing tasks by following guidance or consulting resources
- **Foster** an inclusive, collaborative learning environment and responsible conduct
- **Examine & troubleshoot** commands and code
- **Design & implement** new applications through tinkering and thoughtful design
- **Introspect** on learning and know when to seek help

What is this course?

- The **first course** in the Information Sciences **major and minor**.
- A core, hands-on course that explores different **facets of computing**.
- For students in their **first semester** with limited prior exposure to computing.
- Built around **activities and projects** rather than lectures.
- Goal — move students from **passive consumers** to **active creators** and **problem solvers**.

Facets of computing

- **Command line** — become comfortable driving a computer through a command line interface and the Linux shell.
- **Programming** — use a general-purpose programming language (Python) to build structured tools.
- **Microcontrollers** — interface a computer with the external world using ESP32 hardware.

Intended Learning Objectives

Five objectives guide what you will be able to do at the end of the course.

Learning Objectives

- **ILO 1 • Execute** — carry out computing tasks by following guidelines or consulting manuals.
- **ILO 2 • Collaborate** — foster an **inclusive**, responsible environment and a culture of sharing and care.
- **ILO 3 • Troubleshoot** — infer and explain outputs of commands or code; fix failures.
- **ILO 4 • Create** — apply familiar tasks in creative, thoughtful ways to generate new outcomes.
- **ILO 5 • Reflect** — introspect on learning and know when and how to seek help.

The Three Units

Three explorations, each a facet of computing – command line, Python, microcontrollers.

Course at a glance

Unit	Name	Weeks
Unit 1	Computing with the command line	5
Unit 2	Exploring Python	5.5
Unit 3	Interfacing with Microcontrollers	4.5

Unit 1

Computing with the Command Line

Unit 1 • Command line (5 weeks)

- Explore **vignettes of computing history**.
- Learn to use a **command line interface**.
- Use course-specific tools to **create and update web pages** with sketches.
- Write **simple shell scripts** to automate and combine manual steps.
- Appreciate the utility of the **Linux shell**.

Unit 2

Exploring Python

Unit 2 • Python (5.5 weeks)

- Move from the command line to a **general-purpose programming environment**.
- Learn the **basics of structured programming** in Python.
- Build **more complex tools** than in Unit 1.
- Understand what happens **behind the scenes** when tools run.

Unit 3

Interfacing with Microcontrollers

Unit 3 • Microcontrollers (4.5 weeks)

- Interface your computer to the **external world**.
- Use **ESP32 microcontrollers** programmed with **MicroPython**.
- Work with **sensors and output devices**.
- Realize that computation **need not always be abstract**.

How You'll Learn

**In-class activities, projects and groups —
structured for many kinds of learners.**

How you'll learn & supported

- **In-class activities** & project-based learning form the core.
- **Group work** is an essential part of the course.
- **Diversity** is addressed through group work and **peer tutors / associates**.
- Reasonable support is available for formal **academic accommodations**.

Assessments

A balanced set of assessments map to the intended learning objectives.

Assessment breakdown

Assessment	Count	Weight (each)	Overall
In-class Reflections	2	4% + 7%	11%
Group Presentation	1	5%	5%
Group Projects	3	5 + 10 + 10%	25%
Activity Reports	best 3	3%	9%
Test	1	20%	20%
Learning Journal	1	20%	20%
Class Participation	1	10%	10%

What each assessment looks like

- **In-class Reflections** — write at start (wk 2) and end (wk 14) about views of computing.
- **Group Presentation** — explore a vignette of computing history; an early chance to know classmates.
- **Group Projects** — open-ended project per unit (groups of 3-4).
- **Activity Reports** — brief report each session, best three submitted.
- **Test** — in-class, predict outputs of commands / Python code (weeks 11-12).
- **Learning Journal** — online journal built from Markdown; six

Resources

- **History:** Ignotofsky 2022 · *The History of the Computer*
- **Shell:** Linux Bash Commands, *Pipes and Filters*, Software Carpentry, *The Linux Command Line*
- **Python:** *Big Book of Small Python Projects*, *Think Python*, *Python Cookbook*
- **ESP32:** *Programming ESP32: Learn MicroPython Coding and Electronics*

Let's Pause

Any questions so far?

Thank You

Any questions?